

Topological State Sums and String Nets

Introduction

Phases of matter are traditionally classified according to symmetry breaking. This is done by identifying local order parameters, with zero expectation value, that attain non-zero expectation value when a symmetry of the Hamiltonian is broken by the system's ground state. For example, the ground state magnetisation of a spin chain attains non-zero expectation value when the $\mathbb{Z}_2$ -symmetry of the Hamiltonian is broken. To name a quantum example, consider the transverse field Ising model which is described by the Hamiltonian

$$H = -J \sum_{\langle ij \rangle} Z_i Z_j - h \sum_i X_i,$$

with J setting the energy scale for interactions between neighbouring spins, $\langle ij \rangle$, and h playing the role of an external magnetic field. The Z_i and X_i are Pauli Z and X operators (respectively) which act on the i^{th} spin. When the coupling to the external magnetic field is much stronger than the pairwise coupling, $h \gg J > 0$, the ground state is a tensor product of X_i -eigenstates with eigenvalue $+1$; $|\text{ground state}\rangle \sim \bigotimes_i |x_i = 1\rangle$. The expectation value of the local order parameter, $\langle Z_i \rangle$, is zero in this case. This corresponds to a 'disordered phase' of the ground state which respects the $\mathbb{Z}_2$ -symmetry of the Hamiltonian; $Z_i \mapsto -Z_i$. By contrast, when the pairwise coupling is dominant, $J \gg h > 0$, the $\mathbb{Z}_2$ -symmetry of the Hamiltonian is spontaneously broken by the ground state. In this instance there are two orthogonal ground states, which are the two aligned spin configurations, $|\text{ground state}\rangle_{\pm} \sim \bigotimes_i |z_i = \pm 1\rangle$. The local order parameter $\langle Z_i \rangle$ is now generically non-zero, which signals that the Hamiltonian's $\mathbb{Z}_2$ -symmetry is no longer respected. This state of affairs is referred to as the ground state's 'ordered phase'.

In the above discussion we see that the ordered phase of the transverse field Ising model is degenerate (it is a linear superposition of two aligned states). Naturally, one wants to identify quantum numbers which lift the degeneracy of the ordered phase. In this case, any local $\langle Z_i \rangle$ measurement will project onto one of the two aligned states $\bigotimes_i |z_i = \pm 1\rangle$. There are, however, examples of quantum systems with ground state phases whose degeneracy is not lifted by the expectation value of any local operator. Orthogonal states of such phases are instead distinguished by the expectation values of operators which are manifestly non-local. Operators which are agnostic of local features necessarily measure topological properties of the system. Any ground state phase whose degeneracy is lifted by probing topological information is called a 'topological phase' and such systems are said to have 'topological order'. The simplest lattice model with topological order is the toric code, which was introduced by Kitaev in [1].

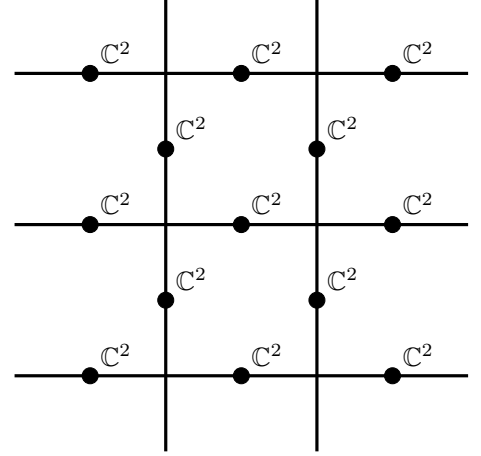
To recapitulate, the quantum numbers of topological phases are robust against local perturbations of the system. From the standpoint of quantum computing, this appears as a platform to store quantum information in a way which is resilient against thermal fluctuations (at least for fluctuations of order similar to the energy gap separating the first excited state from the ground state). For a full quantum computing toolkit, we then require some means to enact a universal gate set on these quantum numbers (i.e. a family of unitary operators which compose to arbitrary unitary transformations). In two-dimensional quantum systems, gate sets are furnished by the manipulation of low energy quasiparticle excitations – the so-called 'anyons'. Quantum computation by anyons is a topic studied thoroughly in the quantum computing community (for example, in Kitaev's paper [2]) and is a focal point of this essay.

To begin, we'll study the toric code as a topological phase and classify its low-energy excitations as quasiparticle species. The properties of the toric code's quasiparticle excitations are studied in particular (for example, their self- and mutual-statistics) and the gate set they generate is identified. In the second section we study models giving rise to general quasiparticle excitations and scrutinise their underlying algebraic and categorical structure. Finally, we consider the so-called 'quantum double' models, which generalise Kitaev's toric code to a class of models with arbitrary algebraic structure at each lattice site. In doing so, we expost links between the models' algebraic input, ground state degeneracy and anyonic excitations.

1 The toric code

The toric code, introduced by Kitaev in 1997 [1], is a simple local lattice model that is topologically ordered; certain global properties of the system (such as the ground state degeneracy) are fixed by topological invariants of the underlying lattice. The model was conceived as a platform for fault-tolerant quantum computation and, in recent years, key aspects have been verified in a laboratory setting [3, 4].

Figure 1: Our model has qubits (each contributing a factor of $\mathbb{C}^2$ to the Hilbert space) attached to the edges of a rectangular lattice. Opposite sides of the lattice are identified to give a toroidal topology.



To set up the model, imagine a $k \times k$ lattice which has a rectangular connectivity structure (as shown in fig 1). Moreover, assume the lattice has the topology of a torus. Now imagine that on each edge of the lattice lives a qubit with individual Hilbert space $\mathbb{C}^2$. The total Hilbert space of the system is therefore isomorphic to $(\mathbb{C}^2)^{\otimes 2k^2}$ (there are $2k^2$ edges for a torus with k^2 vertices, each having four edges). The dynamics of the system are governed by the Hamiltonian

$$H = -J_A \sum_v A_v - J_B \sum_p B_p, \quad (1.1)$$

where v labels the vertices of the lattice, p labels the ‘plaquettes’ (or faces) of the lattice and $J_{A,B}$ set the energy scale of the star and plaquette terms. The A_v are so-called ‘star’ operators which are products of Pauli X operators acting on edges incident to vertex v . The B_p are so-called ‘plaquette’ operators and are products of Pauli Z operators acting on edges bounding plaquette p [2]

$$A_v = \prod_{i \in \text{star}(v)} X_i, \quad B_p = \prod_{i \in \partial(p)} Z_i, \quad (1.2)$$

where $\text{star}(v)$ refers to the edges incident on vertex v and $\partial(p)$ refers to the edges bounding plaquette p . Clearly $A_v^2 = I$ and $B_p^2 = I$. The A_v and B_p commute for all pairings of v and p . To see this, first recall the commutation rule for tensor products of Pauli operators

$$\prod_{i \in M} X_i \prod_{j \in M'} Z_j = (-1)^{|M \cap M'|} \prod_{j \in M'} Z_j \prod_{i \in M} X_i, \quad (1.3)$$

where M and M' are edge sets defining the operators $\prod_{i \in M} X_i$ and $\prod_{j \in M'} Z_j$; $|M \cap M'|$ is the number of edges shared between the relevant operators. Because any pair of star and plaquette operators share exactly zero or two edges (cf. fig 2) we see they must commute with one another and therefore, with the Hamiltonian, H . Because A_v , B_p and H are simultaneously diagonalisable, we can use the star and plaquette eigenvalues as quantum numbers to label the system’s energy eigenstates (we label these a_v and b_p respectively). Of course, the $\{a_v\}$ and $\{b_p\}$ are not independent (a remark we return to later).

We can, in fact, make an even stronger statement about commutativity of star and plaquette operators. Appealing to the visualisation of fig 2, it's clear that any product of stars over a vertex set V , $\prod_{v \in V} A_v$, commutes with any product of plaquettes over a plaquette set P , $\prod_{p \in P} B_p$, because the boundaries of such tilings always share an even number of edges¹. Hence, arbitrary products of A_v and B_p operators are simultaneously diagonalisable with the Hamiltonian and with one another; therefore their eigenvalues provide no new quantum numbers with which to label the states.

There is a topological argument which illustrates why tilings of stars and plaquettes produce no new quantum numbers. Supposing we have a tiling of A_v s or B_p s, we can always deform the tiled operator into a lone A_v or B_p via multiplication with stars or plaquettes (recall $A_v^2 = I$ and $B_p^2 = I$). Therefore, constructing X (respectively Z) loops in the lattice using star (respectively plaquette) tilings creates no new quantum numbers with which to label the system's energy eigenstates (their eigenvalues are just products of the eigenvalues of the component loops). In this sense all contractible loops of the toric code can be equivalence-classed.

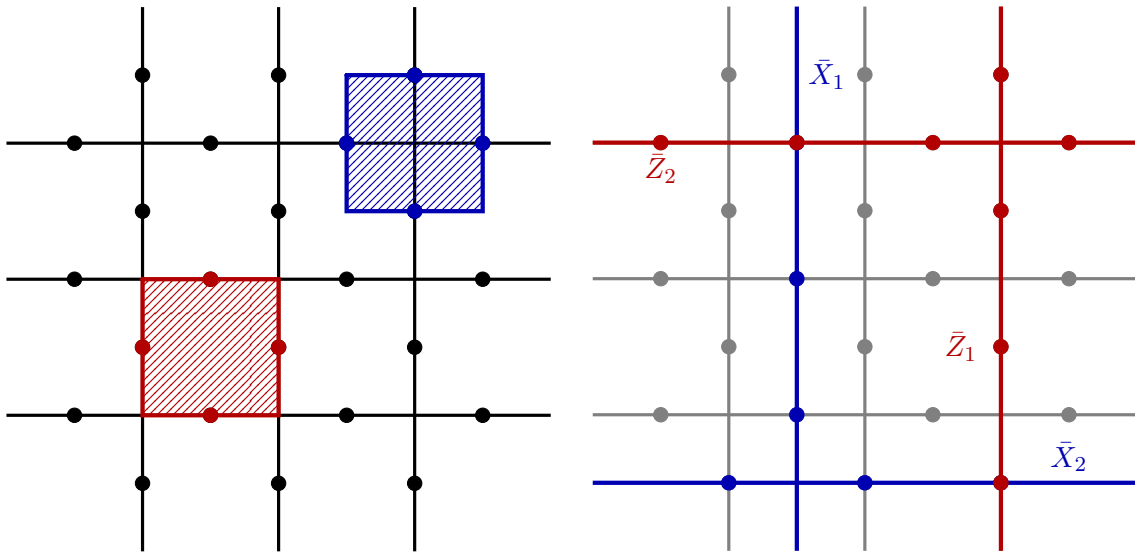


Figure 2: We visualise a Pauli X operator on edge i with a blue line orthogonal to that edge. A Pauli Z operator on edge i is drawn as a red line, parallel to the respective edge. Representative star and plaquette operators are shaded in blue and red respectively on the left. On the right, we show representative non-contractible loops of the torus. In blue are two topologically distinct, non-contractible loops created using X operators. In red are the corresponding loops but created using Z operators. The resulting loop operators are not expressible as tilings of the stars or plaquettes pictured on the left.

New integrals of motion are obtained by leveraging the lattice topology; placing lines of consecutive Pauli X or Z operators along the non-contractible loops of the torus creates operators which commute with the Hamiltonian but not with one another – providing new quantum numbers for our states. Specifically, one can imagine a string of Pauli X (or Z) operators are placed along non-contractible loops of the lattice and its dual (as shown in fig 2). Up to deformation using stars or plaquettes, there are two non-contractible loops on the torus, which we could index with $i = 1, 2$. Let the string of Pauli operators along the i^{th} non-contractible loop of the torus be denoted by $\bar{X}_i$ or $\bar{Z}_i$ respectively (as shown in fig 2).

Because each non-contractible loop operator, $\bar{X}_i$ or $\bar{Z}_j$, intersects any star or plaquette operator exactly

¹Recall that the Pauli operators are involutory. Therefore edges residing inside star or plaquette tilings act with the identity (exactly two stars or plaquettes share any internal edge, so the respective Paulis square). Only the boundary edges carry nontrivial operators.

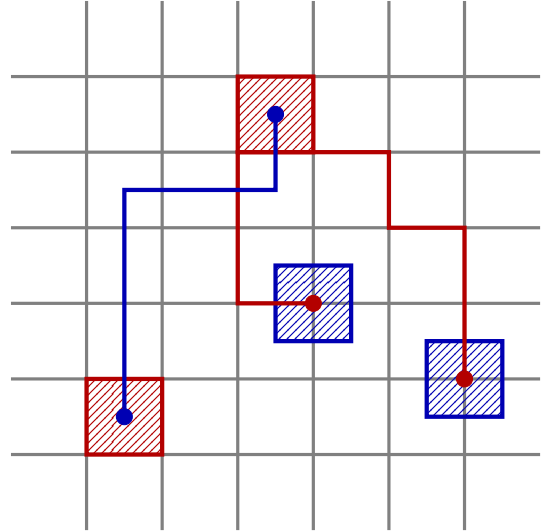
zero or two times, it follows from (1.3) that $[\bar{X}_i, A_v] = [\bar{Z}_i, A_v] = [\bar{X}_i, B_p] = [\bar{Z}_i, B_p] = 0$ hence $[\bar{X}_i, H] = [\bar{Z}_i, H] = 0$; the eigenvalues of $\bar{X}_i$ and $\bar{Z}_i$ can be used to label energy eigenstates. However, the operators $\bar{X}_i$ and $\bar{Z}_j$ have non-trivial commutation relations. It's clear that $[\bar{X}_i, \bar{X}_j] = [\bar{Z}_i, \bar{Z}_j] = 0$. Fig 2 clarifies that, up to homotopy, $\bar{X}_i$ intersects $\bar{Z}_j$ an even number of times when $i = j$ and an odd number of times when $i \neq j$. Hence, (1.3) implies

$$\bar{X}_i \bar{Z}_j = (-1)^{1-\delta_{ij}} \bar{Z}_j \bar{X}_i. \quad (1.4)$$

Hence, the set of operators $\{\bar{X}_1, \bar{X}_2, \bar{Z}_1, \bar{Z}_2\}$ is not simultaneously diagonalisable. We could, for example, choose just $\{\bar{X}_{1,2}\}$ or $\{\bar{Z}_{1,2}\}$ to provide our quantum numbers (in addition to the $\{a_v\}$ and $\{b_p\}$). Importantly, the energy of the state is independent of the $\bar{X}_i$ (respectively, $\bar{Z}_i$) eigenvalues. Therefore the ground state is *at least* four-fold degenerate. Recall there are k^2 star operators (one for each vertex) and k^2 plaquette operators (one for each face). Every edge in the lattice appears in two star operators (those centred on the endpoints of the given edge) and two plaquette operators (those sharing the given edge). It's then clear that $\prod_v A_v = I$ and $\prod_p B_p = I$ because each edge appears in exactly two stars and plaquettes and the Paulis are involutory. Therefore, there are $2k^2 - 2$ independent stabilisers (operators which leave the ground state invariant). The ground states live in the $+1$ -eigenspace of all stabilisers. For m independent and commuting stabilisers on N qubits, the joint $+1$ -eigenspace has dimension 2^{N-m} . This is because each stabiliser squares to the identity and its ± 1 -eigenspaces have equal dimension (projecting onto $+1$ halves the space and stabiliser independence ensures the projections compose). The ground state subspace, $\mathcal{H}_{GS}$ then has dimension $\dim \mathcal{H}_{GS} = 2^{2k^2 - 2k^2 + 2} = 4$ which corresponds exactly with the four-fold degeneracy described by the non-contractible loop operators. On a closed orientable surface of genus g , the analogous counting gives $\dim \mathcal{H}_{GS} = 4^g$ – reflecting the existence of $2g$ independent non-contractible loops on the surface [1]. The ground state degeneracy is therefore a topologically-determined property of the model.

Of practical interest to us are the low energy excitations of the toric code. Understanding that ground states live in the eigenspace with the a_v and b_p equal to $+1$, excitations of the system must have some a_v or b_p eigenvalues flipped to -1 . Formally, this is achieved by acting on a ground state with Pauli X_i or Z_i operators (more generally any product of X or Z operators that do not form loops in the lattice; $\prod_{i \in \alpha} X_i \prod_{j \in \beta} Z_j$ where α and β are unions of curves with endpoints in the lattice). Representative strings of X and Z operators acting on a section of the lattice are shown in fig 3.

Figure 3: In blue is a representative open string of X operators. In red is a representative open string of Z operators. The endpoints of the X (respectively Z) strings intersect individual plaquettes (respectively stars) exactly once, which implies the eigenvalues of those stabilisers are flipped to -1 (we assume the string operators act on a ground state). The excited plaquettes and stars are shaded. Stabilisers which lie along the ‘interior’ of a string maintain unchanged eigenvalue; the string shares exactly two edges with the stabiliser and therefore commutes by (1.3).



Knowing that X_i serves to flip the Z_i eigenvalue at site i (and vice versa) it's easy to see that the eigenvalue of A_v is flipped by any Z -string ending at vertex v and that the eigenvalue of B_p is flipped by any X -string ending in plaquette p . To illustrate, suppose that $\prod_{i \in \alpha} Z_i$ terminates on some vertex v . Then, (1.3)

implies $A_v \prod_{i \in \alpha} Z_i |\text{ground state}\rangle = - \prod_{i \in \alpha} Z_i |\text{ground state}\rangle$. We know $A_v |\text{ground state}\rangle = |\text{ground state}\rangle$ because A_v is a stabiliser. Hence, $\prod_{i \in \alpha} Z_i |\text{ground state}\rangle$ has A_v eigenvalue $a_v = -1$, as asserted. There are two such flipped star eigenvalues associated to each Z -string, which require an energy $4J_A$ to create. An open string will always have two endpoints on a closed lattice (cf. fig 3). Moreover, all configurations of X or Z operators without loops can be expressed as a union of simple open strings, each of which have two endpoints. For example, a simple ‘T’ configuration is really two strings with four endpoints (there is an endpoint at the vertex of the ‘T’). The excitations associated to open string operators are manifestly local and can be moved around the lattice through successive application of more string operators (in this sense they behave like particles). Excitations created by Z -strings are called e quasiparticles (while those created by X -strings are called m quasiparticles). Immediately, we see that the toric code is gapped, with the first excited state having energy $4 \min(J_A, J_B)$.

One might wonder about the statistics of e and m quasiparticles. In the context of surface code excitations the term ‘particle statistics’ adopts a double meaning. In the first sense, it might refer to the so-called ‘topological spin’ of the particle; the phase obtained by the state upon a 2π rotation of the particle. In the second sense, ‘statistics’ can refer to the phase obtained by the state when two particles of specified types exchange places. For the moment, we’ll focus on the so-called ‘exchange statistics’ and return to topological spin in a later discussion.

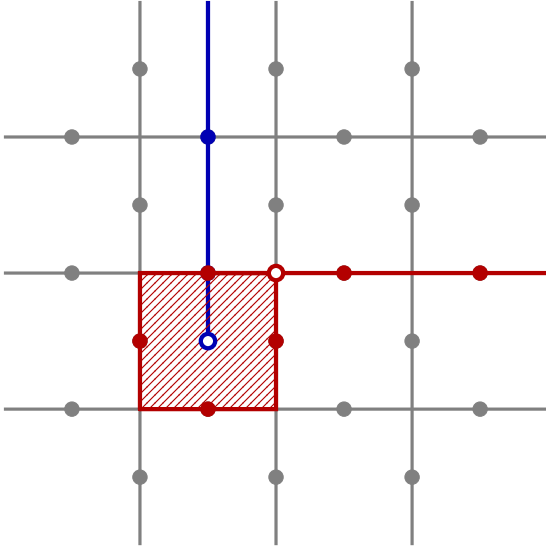


Figure 4: This diagram illustrates how an e charge is looped around an m charge. Again, X operators at sites i are visualised as blue lines orthogonal to the relevant edge. Z operators are visualised as red lines parallel to the relevant edge. The endpoints of open strings giving rise to m (respectively e) excitations are indicated with hollow circles with blue (respectively red) outline. To encircle the e around the m we act with a single B_p (in this case).

To illustrate, we’ll compute the exchange statistics of an e quasiparticle with an m quasiparticle. Imagine there are e and m excitations nearby one another in the lattice (cf. fig 4). Now consider looping the e charge around the m charge (formally achieved by acting with a tiling of B_p operators whose boundary describes the path of the e). The original state is something like $\prod_{i \in \alpha} Z_i \prod_{j \in \beta} X_j |\text{ground state}\rangle$ where α and β are open strings in the lattice whose endpoints give rise to e and m excitations (respectively). To encircle the m with the e , we act with an operator of the form $\prod_{i \in P} B_p$ with P being an appropriate plaquette set. Because the boundary of $\prod_{p \in P} B_p$ intersects β an odd number of times, (1.3) implies

$$\prod_{p \in P} B_p \prod_{j \in \alpha} Z_j \prod_{k \in \beta} X_k |\text{ground state}\rangle = - \prod_{j \in \alpha} Z_j \prod_{k \in \beta} X_k |\text{ground state}\rangle, \quad (1.5)$$

where all we’ve done is commute the B_p -tiling all the way to the right in order to act it on the ground state. Therefore, we see that the state obtains a phase -1 after the e fully encircles the m . Since a full encircling of one particle around another corresponds to two transpositions, we identify the phase obtained due a single exchange as (up to a sign) the root of the encircling phase. For an e and m exchange this evaluates to $\pm i$.

The exchange phase, in contrast to the usual Fermi/Bose dichotomy, is not ± 1 for an e and m exchange. Accordingly, we refer to e and m as ‘mutual semions’. This situation is analogous to the Aharonov-Bohm phases obtained by electrons which loop around magnetic fluxes; first described on theoretical grounds by Aharonov and Bohm in [5] and later demonstrated practically by Tonomura et al. in [6].

Similar arguments show that the state obtains no phase when like charges are exchanged. In this instance, two like charges reside at the endpoints of open string operators which are also of the same type. Exchanging the particle positions is formally achieved by acting on the state with a tiling of stabilisers (stars for X -strings and plaquettes for Z -strings) that connects the endpoints of the relevant strings. Since the stabilisers forming the tiling are defined as tensor products of the same Pauli operators forming the strings, they commute trivially and the state picks up a unit phase. Therefore, the e and m charges are bosons.

The more interesting configuration, by far, is the so-called ‘dyon’. The dyon, or ϵ charge, is a composite particle of the toric code. Dyons are manufactured by ‘localising’ e and m charges and manipulating them in unison. Formally, this is achieved by bringing the endpoints of X and Z strings close to one another on the lattice. For example, the setup of fig 4 has an e charge (the endpoint of the Z string) residing on one of the corners of the plaquette hosting an m charge (the endpoint of the X string). This excited vertex-plaquette pair is precisely the dyon excitation.

Unlike the e and m charges, the dyon has obvious internal structure – it consists of an excited plaquette p and vertex v , with the added detail that v is one of the corners of p . Specifying exactly which corner is the excited one is the sense in which we say the dyon has internal structure. Such a construction allows us to define an unambiguous ‘orientation’ of the dyon (simply draw an arrow from the plaquette centre to the excited vertex cf. fig 5). Since dyons can have an orientation, we can also speak rigorously about the notion of ‘rotating’ an ϵ excitation; simply ‘walk’ the e around the edges of the excited plaquette by acting with the relevant Z operators and the dyon’s orientation will be changed. To affect a 2π rotation, one acts with a B_p operator (p the excited plaquette) which is exactly the situation shown in fig 4. The phase obtained by the dyon state upon a 2π rotation coincides with the phase obtained when a lone e encircles a lone m ; -1 . The phases obtained by quasiparticle excitations upon 2π rotations are known as their ‘topological spin’.

The exchange statistics of the dyon with e and m charges are simple to calculate. When an ϵ encircles an e (respectively m) the state picks up a phase -1 . For example, consider looping an ϵ around an e . In this case the ϵ drags X and Z strings around the e charge. The Z -loop commutes with the open string associated with the lone e , contributing no net phase. On the other hand, the X -loop anticommutes with the lone e ’s open Z -string and contributes the net phase -1 . A similar argument shows that the state attains a phase -1 when an ϵ is looped around an m . Therefore, ϵ charges are mutually semionic with lone e and m charges.

The self-statistics of the dyon are its most interesting feature. To begin, imagine there are two ϵ excitations nearby one another in the lattice. For simplicity, we assume both dyons have the same orientation. If the two dyons do not have the same orientation, their exchange implicitly involves a rotation of one particle. The net phase attained by the state after such a transformation therefore includes contributions from pure particle exchange as well as topological spin. To isolate the exchange statistics we maintain that orientation is preserved. Exchange of the two dyons is then formally achieved by acting on the state with X and Z loops that connect the m and e charges of each dyon, respectively.

If the dyons are exchanged in an orientation-preserving manner – which is the procedure illustrated in the lefthand panel of fig 5 – then the net phase attained by the counting the number of intersections between the exchange loops and the open strings which create the e and m charges. Knowing that a factor of $\pm i$ multiplies into the phase on each crossing (the e and m charges are mutual semions) we obtain a net phase for the pure exchange as -1 . Hence, the dyons are fermions (which corresponds with our previous observation that they have topological spin -1).

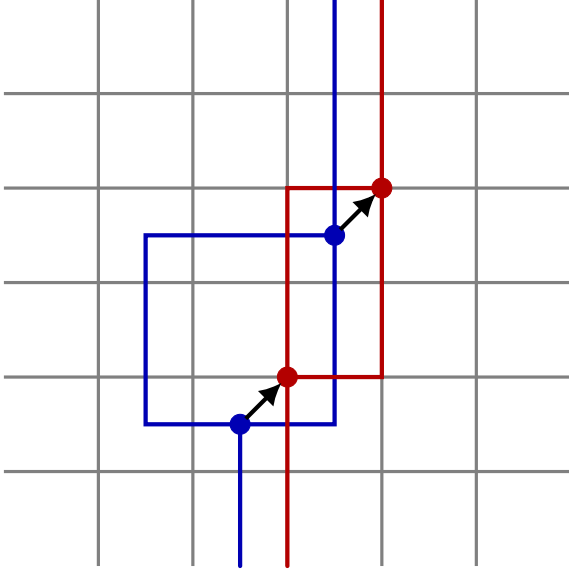


Figure 5: To calculate the self-statistics of the dyon, we picture a representative dyon pair with both dyons having equal orientation. By acting on the state with appropriate X and Z loops connecting the e and m charges of each dyon, we see that there are two intersections with the creation strings; which for a simple exchange – not a full encircling – we know contribute a factor of $\pm i$ each to the state's net phase.

To recapitulate what we've achieved so far, we firstly wrote down a rather simple Hamiltonian (1.1) which was built using local operators A_v and B_p . By putting the underlying lattice on a torus, we classified the quantum numbers which label the system's ground states. In particular, we found that the toric code has four ground states labelled by the eigenvalues of $\bar{Z}_1$ and $\bar{Z}_2$ (alternatively $\bar{X}_1$ and $\bar{X}_2$). The operators $\bar{Z}_1$ and $\bar{Z}_2$ are strings of Pauli Z operators which act along the two non-contractible loops of the torus. The eigenvalues of $\bar{Z}_{1,2}$ are invariant under any homotopy of the underlying loop (stemming from the fact that $[\bar{Z}_{1,2}, B_p] = 0$). In this sense, the information encoded by the $\bar{Z}_{1,2}$ eigenvalues is 'topologically protected' from the action of local operators cf. fig 6. A similar statement is true regarding the eigenvalues of $\bar{X}_{1,2}$.

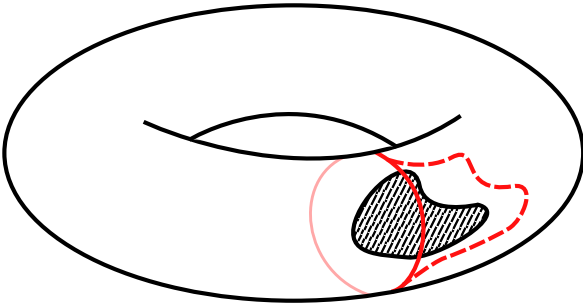


Figure 6: Here is a representation of the toroidal lattice whereupon the Hamiltonian (1.1) lives. In solid red is the non-contractible loop operator, $\bar{Z}_1$ (cf. fig 2) and the shaded region is the support of some local operator. Because the support of $\bar{Z}_1$ is defined up to homotopy, we can always deform the loop operators away from any local disturbances (e.g. the dashed red line). Therefore, the eigenspectra of the $\bar{Z}_{1,2}$ are 'topologically protected'.

The toric code's ground states being characterised by topologically-protected parameters exemplifies topological order. From the standpoint of quantum computing, the temptation is to store information in the $\bar{Z}_{1,2}$ eigenspaces, which provide bases for two so-called 'logical qubits'. The loop operators $\bar{Z}_{1,2}$ and $\bar{X}_{1,2}$ are realised dynamically by anyon transport; creating a pair of e charges, dragging one around the cycle described by $\bar{Z}_i$ and re-annihilating the pair acts $\bar{Z}_i$ on the state. Similarly, the same process with m charges around the conjugate cycle implements $\bar{X}_{2,1}$, which by (1.4) anticommute with $\bar{Z}_{1,2}$ and therefore acts as a logical Pauli X on the corresponding logical qubit. Altogether, the $\bar{Z}_{1,2}$ and $\bar{X}_{2,1}$ are a presentation of the logical Pauli group (which is not a universal gate set). The logical Pauli group consists of tensor products of the $\{I, X, Y, Z\}$, with phases taken from the fourth roots of unity. In the pursuit of a universal gate set, the rest of this essay will explore more general lattice models which give rise to non-Abelian excitations.

2 Anyons, braiding and fusion

Previously, we saw that low-energy excitations of the toric code display particle-like properties (which motivated us to distinguish three nontrivial ‘quasiparticle’ species: e , m and ϵ). Of particular interest was that transport of e and m charges achieved rudimentary quantum computation on two logical qubits. In this section, we’ll generalise the properties of the toric code’s e and m charges by considering lattice models with arbitrary quasiparticle excitations. By imposing some weak criteria on their exchange and fusion properties we’ll exposit the links between the lattice topology and properties of the system’s Hilbert space.

2.1 Particle exchange and the braid group B_N

We will develop a general theory of quasiparticle statistics in surface code models (which will realise the toric code’s e , m and ϵ charges as a particular example). Firstly, suppose we have a Hamiltonian built from local operators residing on a two-dimensional lattice. Generically, the low-energy excitations of a such a Hamiltonian are quasiparticle excitations. These charges may come in multiple species which we label as a , b , c and so on. Now prepare a state consisting of N charges of like species whose positions are fixed (for simplicity, along a line). Now imagine that two neighbouring charges exchange places with one another. Drawing the particles’ worldlines produces a picture like in fig 7. One might recognise fig 7 as a *braid*

Figure 7: Schematic diagram for the worldlines traced out by three particles being permuted in $2 + 1$ dimensions. In this setup, particles 1, 2 and 3 are arranged along a line in some two-dimensional lattice with particles 2 and 3 being transposed (clockwise). Time runs vertically in this picture. In the braid group B_3 , this diagram corresponds to the generator σ_2 .

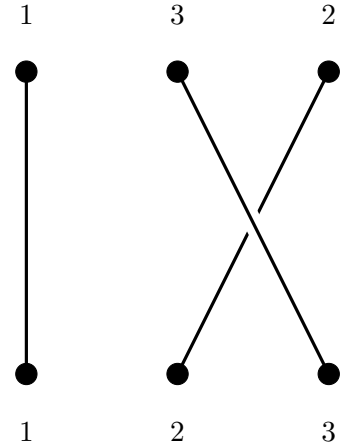


diagram. Indeed, permutations of particles in $2 + 1$ dimensions can generally be understood as braids of the particles’ worldlines. Assuming that our Hilbert space consists of all states with N charges of specified type in fixed positions, the effect of particle permutation must be packaged in unitary representations of the so-called ‘braid group’.

The braid group on N particles, B_N , is the group of isotopy classes of non-intersecting, timelike worldlines in $2+1$ dimensions (with fixed initial and final positions). Two sets of worldlines are equivalent under isotopy if they can be continuously deformed into one another (i.e. without intersection or cutting) while keeping fixed the particles’ initial and final positions. The group structure on B_N arises once we allow composition of braids; to compose two B_N -elements we glue the endpoints of one braid to the starting points of another. Notice then the so-called ‘trivial braid’ (the isotopy class of the set of worldlines which are stationary in configuration space) acts as neutral element. The full braid group B_N is then generated by transpositions of neighbouring particles; we denote the clockwise transposition of particles i and $i + 1$ by $\sigma_i \in B_N$. Inverses σ_i^{-1} are the anticlockwise transpositions of particles i and $i + 1$. It’s necessarily true that [7]

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \quad 1 \leq i \leq N - 2, \quad (2.1)$$

$$[\sigma_i, \sigma_j] = 0 \quad |i - j| \geq 2, \quad (2.2)$$

which are easily verified by drawing the braid diagrams associated to the left- and righthand braidwords of (2.1) and (2.2). Extra constraints may be imposed to reflect the topology of the lattice wherein the particles live. For example, on S^2 there is the additional constraint $\sigma_1 \sigma_2 \dots \sigma_{N-2} \sigma_{N-1}^2 \sigma_{N-2} \dots \sigma_2 \sigma_1 = I$ which reflects the fact that if particle 1 is looped around all other $N - 1$ particles, returning to its original position, the system's trajectory is isotopic to the trivial braid [7] (these are the so-called 'spherical braid groups', cf. fig 8).

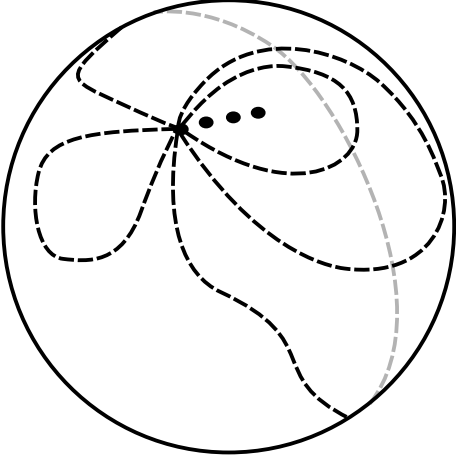


Figure 8: Additional constraints on the B_N -generators might arise as a reflection of the underlying spacetime's topology. For example, the picture on the left shows four particles which live on the surface of S^2 . If the leftmost particle is looped around around the remaining three (orientation doesn't matter in this case) then the trajectory of that particle is contractible to a point on S^2 . Therefore, the corresponding braid group element must equal the identity braid, $\sigma_1 \sigma_2 \sigma_3^2 \sigma_2 \sigma_1 = I$.

Moving to $3 + 1$ dimensions, the group of isotopy classes of N worldlines is the symmetric group S_N . Note that every S_N generator in $3 + 1$ dimensions is involutory; one dimensional worldlines can't form knots in $3 + 1$ dimensions so any transposition is self-inverse regardless of orientation [7]. Therefore, S_N admits only two one-dimensional irreducible representations in $3 + 1$ dimensions; a trivial representation mapping every generator to 1 and a signed representation mapping every generator to -1 . The trivial and signed representations of S_N then furnish the ± 1 exchange phases familiar from Bose and Fermi statistics. Subject to certain reasonable constraints (e.g. particles are point-like and interactions are local) it has been argued that no higher dimensional irreducible representations of S_N in $3 + 1$ dimensions are physically viable [8].

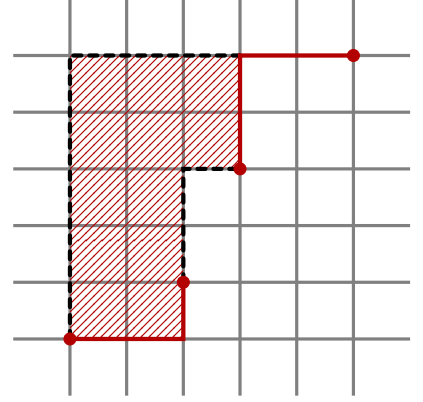
On the other hand, non-trivial generators σ_i of B_N aren't generally self-inverse (which is easy to verify diagrammatically). In this way, the representation theory of B_N is richer than that of S_N (in $3 + 1$ dimensions). For example, the simplest unitary representations of B_N are the Abelian representations, which assign to each B_N -generator a phase, $\sigma_j \mapsto \exp[i\theta_j]$. Because B_N -generators aren't necessarily self-inverse we needn't have $\exp[i2\theta_j] = 1$, allowing arbitrary values for θ_j . In this sense, the exchange phases may differ from the ordinary ± 1 phases of Bose and Fermi statistics. Quasiparticles which realise such arbitrary statistics are called (Abelian) *anyons* (a term coined by Wilczek in [9]). Abelian representations are appropriate when the state is specified entirely by the particles' coordinates (up to a phase). Higher dimensional representations of the braid group, which are generically non-Abelian, are necessary when the particles have internal degrees of freedom. As regards quantum computing, the braiding of so-called 'non-Abelian anyons' typically realises a richer gate set than the braiding of Abelian anyons (e.g. those of the toric code).

To see the connection to the toric code, we can enlarge the braid group by increasing the number of particle species being permuted (i.e. we extend B_N to include generators $\mu_{em,i}$, corresponding to a clockwise exchange of an e charge at site i with an m charge at site $i + 1$). Thinking back to the e and m excitations emerging in the toric code, we saw the double-exchange μ_{em}^2 multiplied our state by an overall phase -1 (meaning that the e and m quasiparticles furnished an Abelian representation of the braid group with $\exp[2i\theta_{em}] = -1$). So we may fix a principal value $\theta_{em} = \pi/2$. In this sense, we see that the mutual statistics of the e and m charges emerges as a consequence of the braid group representation these particles furnish and in particular the fact that $\mu_{em}^2 \neq \text{id}$.

2.2 N -matrices

Keeping our attention on the toric code, we motivate more intricate operations on the e and m charges than just braiding. In §1, we described how two fundamental excitations of the toric code, one e and one m , could be brought together to form a new particle type – the dyon, ϵ . Of course, the pairings of charges we studied in §1 was not exhaustive. For example, we did not consider how two e (respectively m) charges behave when they are brought onto the same site (i.e. ‘fuse’). These cases are actually trivial; for any pair of e excitations can always be regarded as being connected by the same Z -string cf. fig 9. Bringing the two e charges onto the same vertex connects two endpoints of the Z string and forms a loop, equivalent to a tiling of stabilisers. Therefore, in closing the Z -string we have gone from a state with at least two excited vertices to a state with energy reduced by $4J_A$. A similar argument holds for the m charges. In this sense the e and m are their own ‘antiparticles’. Conventionally, we treat such particle-antiparticle composites as being equivalent to a ‘vacuum charge’ which is commonly denoted I [7]. The vacuum has trivial statistics.

Figure 9: By now it’s well-understood that any pair of lattice excitations reside at the endpoints of open X or Z strings. Generally, the string configurations connecting charges are equivalent up to homotopy (deformation with stabilisers). For example, we show four e charges that are, for example, connected by the Z -strings shown in red. By tiling the shaded region with B_p operators, the red Z -strings are mapped to the identity and the charges are instead connected by Z -strings shown in (dashed) black. These two pictures are equivalent as they’re related via a stabiliser tiling.



Understanding that the e and m annihilate themselves, we see there are no composite particles which can be constructed from the basic toric code excitations (e and m) besides the dyon, ϵ . For example, if a dyon ϵ consists of an e at vertex v which corners plaquette p , bringing another e to v causes an annihilation and reduces the dyon to a single m on plaquette p . Similarly, ϵ will fuse with m to produce e . The vacuum has trivial fusion behaviour. The various anyon species of the toric code and their fusion behaviour are shown in table 1. Information contained in fusion tables like table 1 can be neatly packaged into so-called

Table 1: Fusion table for the toric code.

	I	e	m	ϵ
I	I	e	m	ϵ
e	e	I	ϵ	m
m	m	ϵ	I	e
ϵ	ϵ	m	e	I

‘fusion multiplicities’ or ‘ N -matrices’. The N -matrices are defined as three-index objects N_{ab}^c counting the number of fusion channels whereby a and b may fuse into c . For simplicity, we assume that $N_{ab}^c \in \{0, 1\}$ throughout this essay. To be explicit, we can write down the $[N^a]_{ij}$ for the toric code as 4×4 matrices where the indices i and j correspond to the different anyon species of table 1. If we identify $1 \sim I$, $2 \sim e$,

$3 \sim m$ and $4 \sim \epsilon$ then

$$[N^I] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad [N^\epsilon] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$[N^m] = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad [N^\epsilon] = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

which are easily verifiable when compared against table 1. Notably, the N -matrices of the toric code are recognisable as permutation matrices (characteristic of an Abelian theory) [7].

Moving beyond the toric code, we consider a general local lattice model giving rise to anyon species a , b , c and so on. We suppose these different anyons can be brought close to one another and ‘fuse’ into a particle of different type (e.g. particles a and b might fuse to form c ; written $a \otimes b = c$). If more than two particles fuse to a fixed net charge, the composite particle’s long-range properties should be determined only by its charge content. Fusion, which we introduced as a binary operation, must then be associative $a \otimes (b \otimes c) = (a \otimes b) \otimes c$.

We also allow, in general, the existence of multiple fusion channels. For example, a and b might fuse into a net charge c or d (written like $a \otimes b = c + d$). The existence of a vacuum particle I which serves the algebraic role of an identity element is assumed; $a \otimes I = a$ for all a . We also insist on the existence of a unique antiparticle for each charge, say $\bar{a}$ for a , such that $a \otimes \bar{a} \ni I$ (we allow for the possibility that a and $\bar{a}$ have more fusion channels than just I).

Altogether, the rather general conditions laid out in the previous two paragraphs can be formulated as a set of constraints on the model’s N -matrices. Commutativity of fusion implies $N_{ab}^c = N_{ba}^c$. Associativity of fusion means $\sum_d N_{ab}^d N_{dc}^e = \sum_d N_{ad}^e N_{bc}^d$ (i.e. the N -matrices commute). Trivial fusion with the identity implies $N_{aI}^b = \delta_a^b$. Uniqueness of antiparticles means that $N_{ab}^I = \delta_{b\bar{a}}$. A set of anyonic charges with appropriately constrained N -matrices comprise a *commutative fusion ring*.

To remark once more on the formal construction, we start building the theory by selecting a unital $\mathbb{Z}_+$ -ring. $\mathbb{Z}_+$ -rings are finitely generated free $\mathbb{Z}$ -modules whose bases, the so-called ‘ $\mathbb{Z}_+$ -bases’, satisfy $b_i \otimes b_j = \sum_k N_{ij}^k b_k$ with N_{ij}^k non-negative and b_i a basis element. The $\mathbb{Z}_+$ -ring being unital means that the neutral element I , satisfying $I \otimes b = b \otimes I = b$ for all b , must itself be an element of the $\mathbb{Z}_+$ -basis. Associativity of the $\mathbb{Z}_+$ -ring’s multiplication and trivial action of the identity automatically sets the constraints $[N_a, N_b] = 0$ and $N_{aI}^b = \delta_a^b$. By asserting the existence of unique basis elements $\bar{b}$ for every b satisfying $N_{ab}^I = \delta_{b\bar{a}}$, the $\mathbb{Z}_+$ -ring structure is promoted to what’s called a ‘fusion ring’. In practical terms, this amounts to specifying the anyonic content of our theory with a set of appropriately constrained fusion rules.

2.3 The quantum dimension

In lattice models where each site carries a d -dimensional Hilbert space, we understand that the dimension of the whole system’s Hilbert space must scale as a power in d , $\dim \mathcal{H} = d^N$ where N is the number of sites. In this spirit, we imagine our lattice contains a given number, say N , of a charges in fixed positions and consider the Hilbert space of states compatible with this setup, $\mathcal{H}_{N,a}$. The dimension of $\mathcal{H}_{N,a}$ is surely then the number of orthogonal fusion channels which are available to the N copies of a . For an Abelian theory the Hilbert space remains one-dimensional because the particle positions fully specify the

state. Theories which admit non-Abelian representations of the braid group have more degrees of freedom than just the charge positions. Correspondingly, there must be multiple fusion channels available to the anyons; $\dim \mathcal{H}_{N,a} > 1$. The *quantum dimension* associated to the a charges is then defined through the asymptotic relation $\dim \mathcal{H}_{N,a} \sim d_a^N$ which is expected to hold in the large- N limit. In most cases one finds that the quantum dimension obtains non-integer values.

As an example, consider four a charges that fuse to net charge b . The dimension of $\mathcal{H}_{4,a|b}$ is then the number of distinct ways these four a charges can fuse into b . So $\dim \mathcal{H}_{4,a|b} = \sum_{\alpha,\beta} N_{aa}^\alpha N_{a\alpha}^\beta N_{a\beta}^b$. Therefore, $\dim \mathcal{H}_{4,a|b} = [N_a^3]_a^b$. Generically, for N charges we see $\dim \mathcal{H}_{N,a|b} = [N_a^{N-1}]_a^b$ which implies that the dimension of the whole Hilbert space (allowing the net charge to vary) is given as $\sum_b [N_a^{(N-1)}]_a^b$ which gives $\dim \mathcal{H}_{N,a}$ the structure of a trace. Assuming that the N -matrices are diagonalisable, it's then easy to recognise that $\dim \mathcal{H}_{N,a} \sim \Lambda_a^N$ where Λ_a is the largest eigenvalue of N_a . The existence of Λ_a as a non-negative real eigenvalue is guaranteed by the Perron-Frobenius theorem (which states that any non-negative square matrix has eigenvalue of largest absolute value which is unique, real and non-negative).

The N -matrices commute (by associativity of fusion) and are therefore simultaneously diagonalisable. A theorem of Etingof, Gelaki, Nikshych and Ostrik [10] establishes that these matrices admit a common Perron–Frobenius eigenvector $\mathbf{d} = (d_a)$ with non-negative entries, satisfying

$$N^a \mathbf{d} = d_a \mathbf{d}. \quad (2.3)$$

For every anyon type a we identify the components of $\mathbf{d}$ with the quantum dimensions defined earlier. Reading (2.3) component-wise and using $[N^a]_{bc} = N_{ab}^c$ gives

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad (2.4)$$

which expresses the multiplicativity of quantum dimension under fusion; the map $a \mapsto d_a$ is a ring homomorphism from the fusion ring to the positive real numbers.

It is then natural to package the d_a into a global invariant of the theory. We define the *total quantum dimension* as

$$\mathcal{D} = \left(\sum_a d_a^2 \right)^{1/2}, \quad (2.5)$$

where the sum runs over all distinct anyon species (including the vacuum). The quantity $\mathcal{D}$ plays the role of a categorical order; for a quantum double model $D(G)$ (defined in §3) one finds $\mathcal{D}^2 = |G|^2$, in direct analogy with the order of a finite group. As a first example, we compute the quantum dimensions associated with each charge appearing in the toric code. In every case, the largest eigenvalue associated with the N -matrices $N^{I,e,m,\epsilon}$ is $+1$ (in Abelian theories the N -matrices are permutation matrices and their eigenvalues are the corresponding roots of unity). The total quantum dimension of the toric code is therefore $\mathcal{D} = \sqrt{4} = 2$, consistent with $\mathcal{D}^2 = |\mathbb{Z}_2|^2$ for the quantum double $D(\mathbb{Z}_2)$ – a result we obtain explicitly in §3.

Another example to consider are the so-called ‘Fibonacci anyons’. In this theory, there are two charges, I (the identity) and τ , with fusion table shown in table 2.

Table 2: Fusion table for Fibonacci anyons.		I	τ
I	τ	I	τ
τ	τ	τ	$I + \tau$

The N -matrices for the Fibonacci theory may be written

$$[N^I] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad [N^\tau] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (2.6)$$

Firstly, we recognise that the N -matrices of this theory are not permutation matrices, unlike those of an Abelian theory. The quantum dimension associated to the vacuum is easily identifiable as 1. For τ particles the characteristic equation determining the eigenvalues of $[N^\tau]$ is $\lambda^2 - \lambda - 1 = 0$ which admits a larger root $\Lambda = (1 + \sqrt{5})/2$. Therefore the quantum dimension associated with τ anyons coincides with the golden ratio (meaning that $\dim \mathcal{H}_{N,\tau}$ grows like the N^{th} term in the Fibonacci progression, hence the model's name). As a sanity check on (2.4), the fusion rule $\tau \otimes \tau = I + \tau$ reads, through the quantum dimensions, $d_\tau^2 = d_I + d_\tau$, i.e. $\varphi^2 = 1 + \varphi$, which is the defining equation of the golden ratio. The total quantum dimension of the Fibonacci theory is $\mathcal{D} = \sqrt{1 + \varphi^2}$; its irrationality is another signature of non-Abelian statistics (and specifically that this theory cannot be obtained as the quantum double of some finite group). Fibonacci anyons are computationally universal (braiding them can approximate any unitary to arbitrary accuracy), which makes them the simplest known non-Abelian theory capable of universal quantum computation.

2.4 F -symbols

Consider three or more charges that fuse to a given total charge. We can visualise the way charges are 'paired up' by drawing diagrams like what's shown in fig 10 (pictures like this are called "fusion trees"). Associativity of fusion implies that, for fusion trees with equal external charges, the dimensions of the corresponding Hilbert spaces must be equal. However, the spaces themselves generically aren't equal; they are only isomorphic. The change-of-basis that takes one between the spaces associated to different fusion trees is encapsulated in the so-called ' F -symbols' of the model.

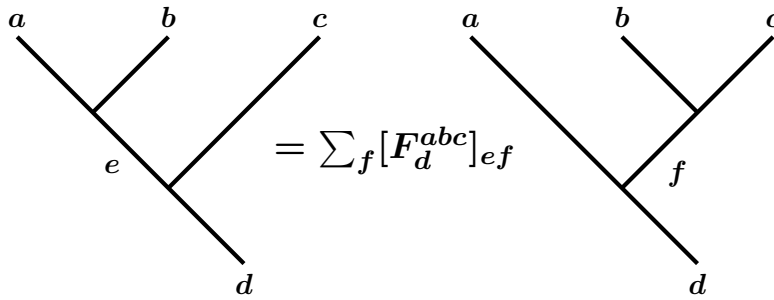


Figure 10: The two fusion trees for $a \otimes b \otimes c = d$. In both cases there is an internal charge labelled with e or f respectively. The diagram on the left represents the fusion $(a \otimes b) \otimes c = d$ while the diagram on the right represents $a \otimes (b \otimes c) = d$. The F -symbol implements the change-of-basis between corresponding Hilbert spaces.

Looking at fig 10, we understand the left- and righthand diagrams represent two different Hilbert spaces, $\mathcal{H}((a \otimes b) \otimes c)$ and $\mathcal{H}(a \otimes (b \otimes c))$ respectively. In both cases, we see that the two trees have one 'free' internal charge indexed by e or f respectively (these indices specify the basis elements in either case). The change-of-basis is then implemented by an F -symbol, $[F_d^{abc}]_{ef}$ which acts like shown in fig 10. Because the F -symbols implement a changes-of-basis between isomorphic Hilbert spaces they are necessarily unitary, $(F_d^{abc})^\dagger F_d^{abc} = I$.

Now, suppose there are four external charges: a , b , c and d fusing to e . In this instance, there are five admissible fusion trees, each of which are related by a change of basis implemented by the appropriate F -symbol. The viable fusion trees are shown in fig 11 and are arranged in a pentagonal configuration where adjacent fusion trees are related via application of a single F -symbol (or ' F -move').

Geometrically, we require that sequences of F -moves starting and ending on the same pair of fusion trees must close (as these represent the same change of basis). By comparing with fig 11, it's clear that closure of equivalent sequences of F -moves implies [7, 11]

$$[F_e^{fcd}]_{gl} [F_d^{abl}]_{fk} = \sum_h [F_g^{abc}]_{fh} [F_e^{ahd}]_{gk} [F_k^{bcd}]_{hl}, \quad (2.7)$$

which is known as the 'pentagon equation'. For any set of fusion rules, we are guaranteed that the pentagon equation admits only finitely-many solutions for the F -symbols (up to gauge equivalence), a property known

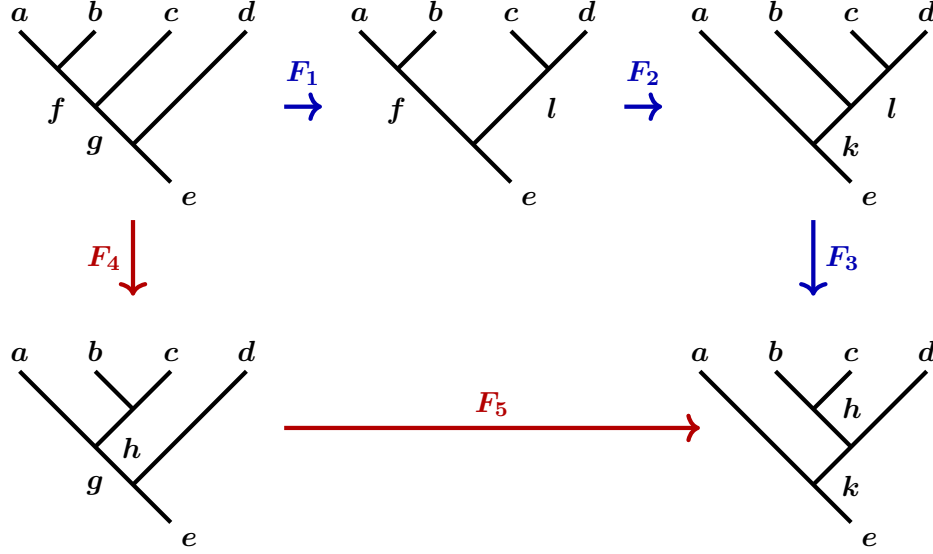


Figure 11: Different viable sequences of F -moves (shown in blue and red) from one basis to another should close; $F_3 \circ F_2 \circ F_1 = F_5 \circ F_4$ (2.7).

as Ocneanu rigidity [11]. Moreover, no new constraints on the F -matrices are generated by increasing the number of incoming charges (guaranteed by MacLane’s coherence theorem [11]).

To put the F -symbols into perspective, we started constructing our theory by enumerating its anyon content (complete with an identity and unique antiparticles for every charge). By specifying the N -matrices (subject to certain constraints) and allowing non-negative formal linear combinations of anyonic charges we obtained a so-called ‘fusion ring’ as the structure describing how anyons can pair up to form new charges. However, there remains an ambiguity as to how we relate the bases of Hilbert spaces associated to different fusion trees (cf. fig 10). Using a simple closure argument (illustrated in fig 11) we found that the F -symbols obey a so-called pentagon equation (2.7) which always admits finitely many solutions (up to gauge equivalence). By choosing a family of F -symbols solving (2.7), we resolve the aforementioned ambiguity and the fusion ring is promoted to a ‘fusion category’.

As a concrete example, we compute the F -symbols associated to the Fibonacci anyons (whose fusion behaviour are summarised in table 2). Clearly, $F_d^{abc} = 1$ whenever one of the incoming charges (a , b or c) is the vacuum I . This is obvious from a diagrammatic standpoint and is illustrated in fig 10 with appropriate substitution into a , b , c and d . F -symbols with two or more incoming charges equal to the vacuum are also trivial. Interesting F -symbols in this case are $F_I^{\tau\tau\tau}$ and $F_\tau^{\tau\tau\tau}$. Looking again at fig 10, we see that $F_I^{\tau\tau\tau}$ is also the identity as there is only one admissible fusion tree (i.e. the associated Hilbert space is one dimensional). To find the components of $F_\tau^{\tau\tau\tau}$ as a 2×2 matrix, we apply (2.7) using $a = b = c = d = e = \tau$ and obtain a system of polynomial equations for the components $[F_\tau^{\tau\tau\tau}]_{ij}$ which are summarised by the following pentagon equation

$$[F_\tau^{f\tau\tau}]_{gl}[F_\tau^{\tau\tau l}]_{fk} = \sum_h [F_g^{\tau\tau\tau}]_{fh}[F_\tau^{\tau h\tau}]_{gk}[F_k^{\tau\tau\tau}]_{hl}$$

Imposing that $F_\tau^{\tau\tau\tau}$ be unitary, the solution for $F_\tau^{\tau\tau\tau}$ admitted by the above system of equations is

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}$$

where φ is the golden ratio and we index $I \sim 1$ and $\tau \sim 2$.

3 Quantum double models

The toric code is an easy-to-understand example of a lattice model with topological order. However, because the toric code's anyonic excitations are Abelian, they cannot realise universal quantum computation through braiding and fusion. We are therefore motivated to study models admitting non-Abelian excitations, where braiding and fusion can, in principle, implement a universal gate set. The appropriate generalisation to make is to the so-called 'quantum double models'. Like the toric code, quantum doubles are local lattice models but are equipped with a richer algebraic structure at each lattice site.

To build a quantum double, one first chooses a finite group G (the simplest non-trivial example being $\mathbb{Z}_2$) and constructs $\mathbb{C}[G]$, the group algebra of G defined over $\mathbb{C}$. Put simply, $\mathbb{C}[G]$ is the $|G|$ -dimensional complex vector space where each element of G is associated to a basis vector; the canonical basis for $\mathbb{C}[G]$ is $\{|g\rangle : g \in G\}$. We assert that the canonical basis is orthonormal, $\langle g|h\rangle = \delta_{g,h}$. We are then able to define four kinds of linear operator on $\mathbb{C}[G]$ which is done in [2] as follows

$$\begin{aligned} L_+^g |h\rangle &= |gh\rangle, & T_+^g |h\rangle &= \delta_{g,h} |h\rangle, \\ L_-^g |h\rangle &= |hg^{-1}\rangle, & T_-^g |h\rangle &= \delta_{g^{-1},h} |h\rangle. \end{aligned}$$

Simple arguments show that these operators obey the following commutation relations

$$\begin{aligned} L_+^g T_+^h &= T_+^{gh} L_+^g, & L_+^g T_-^h &= T_-^{hg^{-1}} L_+^g, \\ L_-^g T_+^h &= T_+^{hg^{-1}} L_-^g, & L_-^g T_-^h &= T_-^{gh} L_-^g, \end{aligned}$$

which will become important when proving exact-solvability of the quantum double Hamiltonian later in this section. The reader might have noticed that the product operators $\{L_+^g T_+^h : g, h \in G\}$, for example, form a linear basis for $\text{End}(\mathbb{C}[G])$, the space of linear endomorphisms on $\mathbb{C}[G]$. To see this, notice that any element of $\text{End}(\mathbb{C}[G])$ is expressible as a $|G| \times |G|$ matrix taking entries in $\mathbb{C}$. The ij^{th} matrix element of $L_+^g T_+^h$ is $\langle i | L_+^g T_+^h | j \rangle = \delta_{h,j} \delta_{i,gh}$ which is one when $(i,j) = (gh,h)$ and zero for every other entry. Hence, the $\{L_+^g T_+^h : g, h \in G\}$ are obviously linearly independent (for distinct g and h) and span $\text{End}(\mathbb{C}[G])$.

As a simple example, we can set $G = \mathbb{Z}_2$ and construct the group algebra $\mathbb{C}[\mathbb{Z}_2]$. We define $\mathbb{Z}_2$ in the usual way, its elements are the congruency classes of the integers modulo two, $\mathbb{Z}_2 = \{0, 1\}$, with group operation given by addition modulo two; $0 \oplus 0 = 0$, $0 \oplus 1 = 1$ and $1 \oplus 1 = 0$. The canonical basis of the group algebra $\mathbb{C}[\mathbb{Z}_2]$ is $\{|0\rangle, |1\rangle\}$. Note that, in the canonical basis, we have

$$L_+^1 = |0\rangle\langle 1| + |1\rangle\langle 0|, \quad T_+^i = |i\rangle\langle i|. \quad (3.1)$$

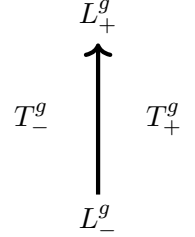
Using (3.1), it's easy to see that L_+^1 is the Pauli X operator, $T_+^0 - T_+^1$ is the Pauli Z operator and that $iL_+^1(T_+^0 - T_+^1)$ is the Pauli Y operator. In other words, starting from $\mathbb{Z}_2$ we have constructed the familiar qubit Hilbert space as the group algebra $\mathbb{C}[\mathbb{Z}_2]$, with Pauli operators being elements of $\text{End}(\mathbb{C}[\mathbb{Z}_2])$ – as expected.

The next step in our construction is to imagine a lattice embedded within an arbitrary (orientable) two-dimensional manifold. As an additional detail which didn't appear in our previous treatment of the toric code, we imagine that each edge of the lattice is *directed*. Then, for an arbitrary finite group G , attach a copy of $\mathbb{C}[G]$ to each edge of the lattice. The states which live on these edges are the so-called 'spins' of the model (each being $\mathbb{C}[G]$ -valued). The full Hilbert space of the model is the tensor product across all edge spaces; $\mathcal{H} = \bigotimes_{\text{edges}} \mathbb{C}[G]$. Changing the orientation of any edge corresponds to a change-of-basis $|g\rangle \mapsto |g^{-1}\rangle$ in that edge's copy of $\mathbb{C}[G]$. We could define the toric code without edge orientations because every element of $\mathbb{Z}_2$ is self-inverse.

Continuing in the fashion of [2], we define translation and projection operators which will build the analogous star and plaquette operators of the quantum double model. To start, select a vertex v and one

incident edge j . The operator $L^g(j, v)$ acts as L_+^g on the j^{th} edge space if the edge j is oriented towards v ; otherwise, $L^g(j, v)$ acts as L_-^g on the j^{th} edge space. Similarly, we can select a plaquette p and one of its bounding edges j . Then the operator $T^g(j, p)$ acts on the j^{th} edge space as T_+^g if p is to the right of j (remember j is oriented); otherwise it acts as T_-^g . In fig 12, we reproduce a diagram of [2] which summarises the edge rules for $L^g(j, v)$ and $T^g(j, p)$.

Figure 12: Suppose the pictured edge is labelled j . If j flows into vertex v , $L^g(j, v)$ acts on the edge state with L_+^g . If j flows out of vertex v , $L^g(j, v)$ acts with L_-^g . If plaquette p is to the left of j , $T^g(j, p)$ acts with T_-^g . If plaquette p is to the right of j , $T^g(j, p)$ acts with T_+^g .



Following [2], we define local ‘gauge transformations’, A_v^g , and ‘magnetic charge’ operators, $B_{(v,p)}^g$

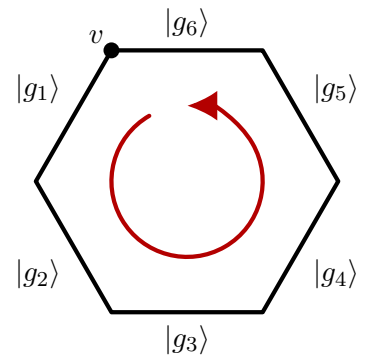
$$A_v^g = \prod_{j \in \text{star}(v)} L^g(j, v), \quad B_{(v,p)}^g = \sum_{g_1 \dots g_k = g} \prod_{m=1}^k T^{g_m}(j_m, p), \quad (3.2)$$

where $j_1, \dots, j_k$ are the bounding edges of plaquette p listed in counter-clockwise order starting from vertex v . The sum in $B_{(v,p)}^g$ is taken over all projectors onto states with holonomy g . If unclear, the holonomy around plaquette p based at v , call it $\text{hol}(p, v)$ is defined to be the ordered product of group elements which live on the bounding edges of that plaquette (conventionally taken in counter-clockwise order). When calculating $\text{hol}(p, v)$, the edge orientations are modified to oppose the direction of traversal (changing basis $|g\rangle \mapsto |g^{-1}\rangle$ on aligned edges). Explicitly, the holonomy around plaquette p will look something like

$$\text{hol}(p, v) = g_1^{\epsilon_1} g_2^{\epsilon_2} \dots g_n^{\epsilon_n}, \quad (3.3)$$

where $\epsilon_i = 1$ for G -elements living on an anti-aligned edge and $\epsilon_i = -1$ for G -elements living on an aligned edge. The factor $g_1^{\epsilon_1}$ comes from the first edge state $|g_1\rangle$ in anti-clockwise traversal around p starting at v ; the factor $g_n^{\epsilon_n}$ comes from the final edge state $|g_n\rangle$ in anti-clockwise traversal around p , ending at v . An example of how the holonomy is calculated for a representative plaquette is illustrated in fig 13. We now

Figure 13: Shown alongside is a hexagonal plaquette p for which we compute the holonomy. The top left vertex v is distinguished as the point at which we’ll base the holonomy. Listed in counter-clockwise order, starting and ending at v , the group elements carried by the plaquette’s edges are g_i with $i = 1, \dots, 6$. If the i^{th} edge has clockwise orientation, we multiply a factor of g_i into $\text{hol}(p, v)$. If the i^{th} edge has counter-clockwise orientation, we multiply a factor of g_i^{-1} into $\text{hol}(p, v)$. Assuming the first three edges are clockwise oriented and the last three counter-clockwise, we’d get $\text{hol}(p, v) = g_1 g_2 g_3 g_4^{-1} g_5^{-1} g_6^{-1}$, for example.



define the plaquette operators as the projectors onto states with trivial holonomy

$$B_p = B_{(v,p)}^e \iff B_p = \delta(\text{hol}(p, v), e). \quad (3.4)$$

Since B_p is diagonal in the group element basis, with eigenvalues $\{0, 1\}$, we identify it as a projector. We elide the base vertex from which the holonomy is calculated, as this is irrelevant when projecting onto trivial

holonomy states (any cyclic permutation of factors $g_1 g_2 \dots g_n$ which multiply to e remains unchanged). To illustrate, suppose the holonomy of a plaquette p calculated at vertex v_1 is neutral, $\text{hol}(p, v_1) = e$. Adopting the labelling convention of fig 13, $g_1 g_2 \dots g_n = e \Rightarrow g_1^{-1} = g_2 \dots g_n$. Supposing we calculate the holonomy at the next vertex along the counter-clockwise traversal of p , call it v_2 , we find $\text{hol}(p, v_2) = g_2 \dots g_n g_1$, which remains e as we've simply commuted g_1 past its inverse. Because the holonomies based at each vertex are obtained by cyclically permuting the group factors of the original holonomy, we see that plaquettes with trivial holonomy at one vertex must have trivial holonomy at every vertex.

When $G = \mathbb{Z}_2$, every group element is self-inverse so the holonomy is trivial whenever an even number of edges carry the state $|1\rangle$. Recalling that $Z = T^0 - T^1$, we see $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$. Therefore product $\prod_{i \in \partial p} Z_i$ has eigenvalue $+1$ whenever the holonomy is trivial and -1 when the holonomy is non-trivial. Hence the projector B_p can be written

$$B_p = \frac{1}{2} \left(I + \prod_{i \in \partial p} Z_i \right), \quad (3.5)$$

which, up to an affine transformation, is the same as the B_p operator we defined for the toric code earlier in this essay. In the language of gauge theory, the B_p operators enforce a 'flatness' condition.

Turning to the primitive star operator A_v^g , we see that it acts on all edges incident on vertex v . Specifically, for edges oriented towards v basis states are mapped $|h\rangle \mapsto |gh\rangle$. For edges oriented away from v , basis states are mapped $|h\rangle \mapsto |hg^{-1}\rangle$. In this way, application of A_v^g commutes with a reversal of the edge orientations; inverting the orientation of an edge replaces $|h\rangle$ with $|h^{-1}\rangle$ and simultaneously swaps the roles of left- and right-multiplication in A_v^g ; the two changes cancel, leaving A_v^g well-defined.

We then define the star operator at vertex v as the average over all gauge transformations A_v^g on that vertex [2, 11]

$$A_v = \frac{1}{|G|} \sum_{g \in G} A_v^g. \quad (3.6)$$

So defined, the A_v acts like a projection operator; $A_v^2 = A_v$. To see this, recognise that $A_v^g A_v^h = A_v^{gh}$ since $A_v^g A_v^h |i\rangle = |(gh)i\rangle = A_v^{gh} |i\rangle$ (on an edge oriented towards v) and $A_v^g A_v^h |i\rangle = |i(gh)^{-1}\rangle = A_v^{gh} |i\rangle$ (on an edge oriented away from v). So then

$$A_v^2 = \frac{1}{|G|^2} \sum_{g,h} A_v^g A_v^h = \frac{1}{|G|^2} \sum_{g,k} A_v^k = \frac{1}{|G|} \sum_k A_v^k = A_v,$$

where we've relabeled $gh = k$ in the second sum (k ranges over all of G as we sum on h ; group translations are surjective). For example, in the toric code A_v^0 acts as the identity on all edge states; $A_v^0 |g\rangle = |0 \oplus_2 g\rangle = |g\rangle$ on edges oriented towards v and $A_v^0 |g\rangle = |g \oplus_2 0\rangle = |g\rangle$ on edges oriented away from v hence $A_v^0 \cong I$. On the other hand A_v^1 flips all edge states; $A_v^1 |0\rangle = |1 \oplus_2 0\rangle = |1\rangle$, $A_v^1 |1\rangle = |1 \oplus_2 1\rangle = |0\rangle$ on edges oriented towards v and similarly for edges oriented away from v . Identifying L^1 as the same as X (3.1), we write $A_v^1 = \prod_{i \in \text{star}(v)} X_i$ and via (3.6)

$$A_v = \frac{1}{2} (A_v^0 + A_v^1) = \frac{1}{2} \left(I + \prod_{i \in \text{star}(v)} X_i \right),$$

which has eigenvalues 0 and 1. This is the projector onto the $+1$ eigenspace of the star operator $\prod_i X_i$ defined in the first section. The two forms of the Hamiltonian — one using the ± 1 stabiliser operators of §1, the other using the $\{0, 1\}$ projectors defined here — differ only by a constant energy shift. We adopt

the projector convention henceforth. The condition $A_v|\psi\rangle = |\psi\rangle$ selects states invariant under all gauge transformations A_v^g at vertex v ; in this sense the star operators enforce local gauge invariance.

Consider now a vertex-plaquette pair (v, p) , with v sitting on the boundary of p . The family of operators $\{D_{(g,h)} = B_{(v,p)}^g A_v^h : g, h \in G\}$ form a linear basis for the so-called 'quantum double' algebra $D(G)$ at the site (v, p) . As remarked in [2], the structure of $D(G)$ is determined by the relations

$$A_v^g A_v^h = A_v^{gh}, \quad B_{(v,p)}^g B_{(v,p)}^h = \delta_{g,h} B_{(v,p)}^h, \quad A_v^g B_{(v,p)}^h = B_{(v,p)}^{ghg^{-1}} A_v^g. \quad (3.7)$$

We have already verified the first equation of (3.7). The second equation follows trivially from the fact that the $B_{(v,p)}^g$ are diagonal in the group element basis. The third equation follows from a straightforward arguments. To begin, notice that if we act with A_v^g , the holonomy of plaquette p (calculated starting at v) is conjugated by g ; $\text{hol}(p, v) = g_1 \dots g_n \mapsto gg_1 \dots g_n g^{-1} = g \text{hol}(p, v) g^{-1}$. To see why, notice that there are four possibilities for how the edges incident on v are aligned; both edges are clockwise-oriented; both edges are counter-clockwise-oriented; both flow into v ; both flow out of v . If both edges are clockwise-oriented, the group element carried by the first edge traversed in the holonomy is modified $g_1 \mapsto gg_1$ while the final edge is modified $g_n \mapsto g_n g^{-1}$ (this follows from simple application of the rules summarised in figs 12 and 13). If both edges are counter-clockwise oriented, the group elements are mapped $g_1 \mapsto g_1 g^{-1}$ and $g_n \mapsto gg_n$ by A_v^g . When calculating the holonomy, the orientation of all edges are implicitly changed to read clockwise, meaning that once again the first factor becomes $g_1 g^{-1} \mapsto gg_1^{-1}$ and the last factor $gg_n \mapsto g_n^{-1} g^{-1}$, which are precisely the first and last factors in the original holonomy, rescaled by g and g^{-1} on the left and right respectively. The other two cases are similar and, in all instances, the action of A_v^g on the appropriately calculated holonomy is to conjugate by g . The third identity of (3.7) follows trivially. As such, the following multiplication rule holds true [2]

$$D_{(h_1, g_1)} D_{(h_2, g_2)} = \delta_{h_1, g_1 h_2 g_1^{-1}} D_{(h_1, g_1 g_2)}.$$

With the star and plaquette operators defined for arbitrary input group G , we write the Hamiltonian [11]

$$H = -J_A \sum_v A_v - J_B \sum_p B_p.$$

Like in the toric code example, the ground states of this Hamiltonian live in the simultaneous $+1$ eigenspace of the A_v and B_p operators. The Hamiltonian remains gapped and, with the new normalisation on A_v and B_p , this gap equals $2 \min(J_A, J_B)$. In this more general class of models, it remains true that $[A_v, A_{\bar{v}}] = 0$, $[B_p, B_{\bar{p}}] = 0$ and $[A_v, B_p] = 0$ for all choices of star or plaquette. Verifying the first two commutation relations is easy; A_v and $A_{\bar{v}}$ commute trivially when they share no common edges otherwise they share exactly one edge and commute because one operator acts by left multiplication while the other acts by right multiplication (operations which always commute); the plaquette operators trivially commute because they are all diagonal in the group element basis.

A more intricate argument is required to show $[A_v, B_p] = 0$. There are two cases to be considered. The first case is when the vertex v does not lie on the boundary of p . In this instance the operators A_v and B_p act on disjoint edge sets and commute trivially. The second case has v on one of the corners of B_p . Without loss of generality, we can imagine B_p as computing the holonomy starting at vertex v (B_p projects onto states of trivial holonomy, which are agnostic of which vertex we start from). The effect of A_v on the holonomy of plaquette p is to average over gauge transformations $\text{hol}(p, v) \mapsto g \text{hol}(p, v) g^{-1}$. Knowing, however, that B_p projects onto states with trivial holonomy sets $g \text{hol}(p, v) g^{-1} = g e g^{-1} = e$ which shows that the holonomy is unchanged. Therefore $[A_v, B_p] = 0$.

The ground states of the quantum double Hamiltonian therefore satisfy $a_v = 1$ at every vertex and $b_p = 1$ at every plaquette. In the language of lattice gauge theory, they are simultaneously gauge-invariant

and flat. Assuming we work within the ground state subspace, every plaquette has trivial holonomy and the holonomy of any closed loop in the lattice can only depend on the loop's homotopy class. To see this, imagine that two neighbouring plaquettes have holonomies $\text{hol}(p_1) = a_1 a_2$ and $\text{hol}(p_2) = b_1 a_2^{-1}$ where a_2 is the group element carried by the shared edge. If $\text{hol}(p_{1,2}) = e$ then $a_1 = a_2^{-1}$ and $b_1 = a_2$. The holonomy around $p_1 \cup p_2$ is then $a_1 b_1 = a_2^{-1} a_2 = e$. Hence a tiling of plaquettes with trivial holonomy itself has trivial holonomy.

The non-trivial case concerns pairs of loops in the homotopy class of a non-contractible cycle (pictured in fig 14). Firstly, we standardise a particular direction to compose the holonomy of a non-contractible cycle; we draw a single non-contractible loop on the torus and assign it an orientation. The direction we compose the holonomy on this reference loop is then continuously mapped to other loops in the same homotopy class. We then imagine two such non-contractible loops on the torus with holonomies g_1 and g_2 (which are calculated at vertices v_1 and v_2 respectively). To compute the holonomy around the region bounded by both loops, we start at v_1 , cycle around the first loop (multiplying a factor of g_1 into the holonomy) and then move from v_1 to v_2 , which will contribute some group element a to the holonomy. Once at v_2 , we cycle around the second loop in a counter-clockwise direction (which, for example, can be specified by the outer-pointing normal of the torus). Cycling around the second loop and returning to v_1 multiplies a factor $g_2^{-1} a^{-1}$ into the holonomy. The net holonomy is then $g_1 a g_2^{-1} a^{-1}$. However, because the loops bound a region which is plaquette-tileable, the net holonomy must be neutral. Therefore $g_1 = a g_2 a^{-1}$; the holonomies of homotopic, non-contractible loops might not be equal but at least belong to the same conjugacy class. As an extra constraint, contractability of the torus' commutator loop implies holonomies around conjugate cycles commute. The system's ground states are therefore labelled by commuting pairs of group elements, which are equivalenced under simultaneous conjugation, $(g_1, g_2) \sim (h g_1 h^{-1}, h g_2 h^{-1})$.

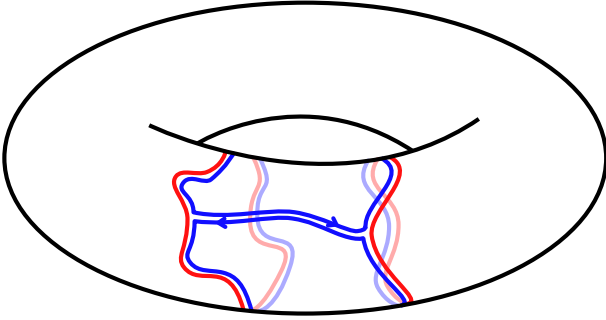


Figure 14: Two non-contractible cycles of the torus are shown in red. To compute the holonomy around the bounded region, we arbitrarily assign an orientation to one (let's say counter-clockwise w.r.t. the bounded region) and assign the same orientation to the other loop. With an orientation, we compute the holonomies around these loops in the usual way. For the bounded region, we connect the two cycles with an arbitrary path (blue) and compute the holonomy in the usual way.

To illustrate some examples, we can look at the quantum doubles built on finite Abelian groups G . Because G is Abelian, every pair in G commutes and the action of conjugation is trivial. Therefore, the ground state degeneracy is $|G|^2$, the number of pairs in G . By specialising to the toric code, which is built on $\mathbb{Z}_2$, we recover our previous result for the ground state degeneracy, $|\mathbb{Z}_2|^2 = 4$. By contrast, the count for non-Abelian groups is not as simple. To find the ground state degeneracy, one must identify the number of orbits (under simultaneous conjugation) associated to each commuting pair in G . This can be done by application of Burnside's lemma, which states that for any finite group G acting on a set X , the number of orbits in X (call it $|X/G|$) is

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\{x \in X : g \star x = x\}|, \quad (3.8)$$

where $\star$ denotes the group action. In our problem, the group G acts by simultaneous conjugation on the set of ordered pairs of commuting G -elements. The summand of (3.8) counts the number of pairs which

are left unchanged under simultaneous conjugation by a particular group element. Therefore, (3.8) can be rewritten

$$\dim \mathcal{H}_{\text{GS}}(G) = \frac{1}{|G|} \sum_{h \in G} |\{(g_1, g_2) \in Z_G(h) \times Z_G(h) : g_1 g_2 = g_2 g_1\}|, \quad (3.9)$$

with $Z_G(h) = \{g \in G : gh = hg\}$ the centraliser of h in G . Because group elements in the same conjugacy class have isomorphic centralisers, (3.9) can be further simplified to read

$$\dim \mathcal{H}_{\text{GS}}(G) = \frac{1}{|G|} \sum_{h^*} |C_G(h^*)| \cdot |\{(g_1, g_2) \in Z_G(h^*) \times Z_G(h^*) : g_1 g_2 = g_2 g_1\}|, \quad (3.10)$$

where h^* are representatives of conjugacy classes in G , and $C_G(h^*)$ is the conjugacy class containing h^* .

As a demonstration, we compute the ground state degeneracy for S_3 on the torus. Firstly, we write the elements of $S_3 = \{e, (12), (23), (13), (123), (132)\}$ where e is the identity, the (ij) are transpositions of i and j and (ijk) are cyclic permutations $i \mapsto j, j \mapsto k, k \mapsto i$. It is known that S_3 has three conjugacy classes which are $\{e\}$, $\{(12), (13), (23)\}$ and $\{(123), (132)\}$. We now count the number of commuting pairs in the centraliser of a representative from each conjugacy class.

The centraliser of e (representative of $\{e\}$) is all of S_3 . The identity permutation commutes with all S_3 -permutations and contributes six commuting pairs to our count. The three transpositions commute with the identity and themselves, contributing six more pairs in total. The cycles each commute with the identity, themselves and the other cycle and therefore contribute another six pairs. The total number of commuting pairs in S_3 is therefore $6 + 6 + 6 = 18$.

The centraliser of (12) (representative of $\{(12), (23), (13)\}$) is $\{e, (12)\} \cong \mathbb{Z}_2$. Since the centraliser is Abelian in this case, all four pairs commute. Similarly, the centraliser of (123) (representative of $\{(123), (132)\}$) is $\{e, (123), (132)\} \cong \mathbb{Z}_3$, which provides another nine pairs to the count.

By summing the number of commuting pairs we found in each centraliser, each rescaled by the size of the corresponding conjugacy class, and then dividing by $|S_3|$ we successfully compute the ground state degeneracy according to (3.10). Specifically, we find $\dim \mathcal{H}_{\text{GS}}(S_3) = (18 + 3 \cdot 4 + 2 \cdot 9)/6 = 8$. The ground state of an S_3 model living on a torus is eight, strictly less than $|S_3|^2 = 36$ due to S_3 being non-Abelian.

It is known that the irreducible representations of $D(G)$ are in bijection with G -orbits of pairs (C, π) , where C is a conjugacy class of G and π is an irreducible representation of the centraliser of a representative of C [2]. The number of such pairs equals the orbit count of simultaneous conjugation on commuting pairs in G (i.e. the ground state degeneracy on the torus computed above). As we show in the next section, these orbits classify the anyonic excitations of $D(G)$. So the number of anyons equals the ground state degeneracy on the torus. Therefore, the counting is linked both to the topology of the underlying manifold (the number of non-contractible cycles) and to the algebraic structure of G (its conjugacy classes and centraliser representations).

3.1 Excitations of quantum double models

By now we're convinced that the system's ground states live in the simultaneous $+1$ -eigenspace of the A_v and B_p operators. Thinking back to our first discussion of the toric code in §1, it's also clear that we produce an excited state by flipping the eigenvalues of some of the A_v or B_p (in that case by acting with open strings of Paulis). Consider now the quantum double $D(G)$ at vertex-plaquette site (v, p) . The eigenspectrum of the B_p operator is determined by the set of all accessible holonomies around plaquette p . In general, the holonomy of a plaquette p transforms like $\text{hol}(p, v) \mapsto g \text{hol}(p, v) g^{-1}$ under gauge transformations A_v^g . Assuming that the anyonic species arising from plaquette violations are invariant with respect to gauge, they must be labelled by the conjugacy classes of G . The conjugacy class is said to measure the *magnetic flux* through a plaquette violation.

Vertex violations, on the other hand, are typically not gauge invariant. Recall that A_v averages over all gauge transformations at vertex v and projects onto the invariant subspace corresponding to the trivial representation of G . If an A_v is excited, it means that certain gauge transformations are no longer trivial. If the conjugacy class associated to the magnetic violation is C , the residual gauge group associated with the electric violation must be the centraliser of C (i.e. we pick a representative of C , say h , and construct $Z_G(h) = \{g \in G : gh = hg\}$). Since A_v^g for $g \in Z_G(h)$ preserves the chosen magnetic flux (i.e. holonomy) representative, h , the A_v^g must act on the internal Hilbert space associated to the flux excitation; i.e. the state transforms under an irreducible representation of the centraliser. There may be different irreducible representations of the centraliser, corresponding to the different kinds of electric charge which may arise at the site [2].

To recapitulate, conjugacy classes, C of G , label the so-called *magnetic flux* through a plaquette violation. The irreducible representation of the corresponding centralisers label the so-called *electric charge*. As an example, consider the case $G = \mathbb{Z}_2$. Since $\mathbb{Z}_2$ is Abelian there are exactly two conjugacy classes, $\{0\}$ and $\{1\}$. Correspondingly, the centralisers of these conjugacy classes are the entirety of $\mathbb{Z}_2$. The group $\mathbb{Z}_2$ has two irreducible representations (which are the trivial representation and the signed representation). The total number of distinct anyon species is therefore $2 \times 2 = 4$, matching the family $\{I, e, m, \epsilon\}$ we classified earlier in this essay. In particular, we can identify the conjugacy class $\{0\}$ as giving rise to two charges (from the trivial representation we obtain I and from the signed representation we obtain e). The conjugacy class $\{1\}$ gives rise to m charges (from the trivial representation) and ϵ charges (from the signed representation).

As another example, consider $G = S_3$, again the permutation group on three elements. As we saw previously, S_3 has three conjugacy classes $C_1 = \{e\}$ with centraliser $Z_{S_3}(C_1) = S_3$, which has three irreducible representations (the trivial, sign, and standard representation); $C_2 = \{(12), (23), (13)\}$ with centraliser $Z_{S_3}(C_2) = \mathbb{Z}_2$, which has two irreducible representations (trivial and sign); $C_3 = \{(312), (231)\}$ with centraliser $Z_{S_3}(C_3) = \mathbb{Z}_3$, which has three irreducible representations (corresponding to the cube roots of unity). The total number of anyon species is therefore $3 + 3 + 2 = 8$. The quantum dimension of an anyon labelled by (C, π) , with C being the conjugacy class and π the irreducible representation of the centraliser, is $d_{(C, \pi)} = |C| \cdot \dim(\pi)$ [11]. Note that the order of the conjugacy class counts the number of gauge-equivalent flux values, while the dimension of the representation counts the internal degrees of freedom of the charge. For the eight anyons of $D(S_3)$ we obtain quantum dimensions $\{1, 1, 2, 2, 2, 2, 3, 3\}$ counted with multiplicity. Those anyons with quantum dimension greater than one are non-Abelian. As a consistency check, the total quantum dimension satisfies

$$\mathcal{D}^2 = 36 = |S_3|^2,$$

validating the relation $\mathcal{D}^2 = |G|^2$ for quantum doubles suggested in §2. The presence of anyons with quantum dimension greater than one in $D(S_3)$ realises non-Abelian topological order (there are multiple fusion channels available to certain anyons and the braid group representation they furnish has dimension greater than one). Whether the resulting gate set is computationally universal is a separate question and is sensitive to the detailed structure of G . For $D(S_3)$ specifically, the gate set is not universal but the construction we've outlined extends to groups whose double does support universal computation. For example, the quantum double $D(S_5)$ admits a computationally universal gate set [2].

Although we don't calculate them, the F -symbols and N -matrices of a quantum double are constrained by the input group G . Details of how these quantities are calculated are exposed in [11, 12]. As a further generalisation of the algebraic framework we've developed in this essay, there exists a class of model where the fusion data is not uniquely determined a priori – the so-called string nets [13, 14]. String nets are built on the unitary fusion categories discussed in §2 which are more general than quantum doubles in that they may allow for different choices of F -symbols on top of the underlying fusion ring.

4 Conclusion

In this essay, we have described the toric code's ground state as a topological phase. In particular, we saw that the ground state degeneracy of the toric code was lifted by operators spanning the non-contractible cycles of the torus. The quantum numbers associated to these operators are resilient against local perturbations and provide a means to store quantum information which is topologically protected. Additionally, the low energy excitations of the toric code were identified as Abelian anyons, whose braiding generate the Pauli group – a non-trivial but also non-universal gate set which acts on the toric code's two logical qubits.

In the pursuit of a universal gate set, we considered models giving rise to arbitrary anyonic excitations, with the hopes that we would encounter and classify non-Abelian anyons. We did exactly this in §2 and found that braiding of non-Abelian anyons furnished representations of the braid group with dimension greater than one. We also discussed in abstract terms the structure of such theories; the anyonic content of any surface code is generically described by a commutative fusion ring, with non-Abelian theories requiring additional specification of the so-called F -symbols (which promote the theory unambiguously to a fusion category). Knowing that gate sets other than the Pauli group are attainable through the manipulation of non-Abelian anyons, we were motivated to study Drinfel'd's quantum doubles (an exactly-solvable class of lattice Hamiltonians which realise non-Abelian excitations).

To construct a quantum double, we imagined the same toroidal lattice of Kitaev's toric code but now carrying a copy of the group algebra $\mathbb{C}[G]$ (for G a finite group) on each edge. The low energy excitations of the quantum double Hamiltonian were identified as quasiparticles, in an analysis similar to what was done for the toric code in §1. The ground state of the quantum double was seen to be degenerate, with the degeneracy lifted only by non-local operators acting along non-contractible loops of the torus (i.e. the ground state of the quantum double on a torus is a topological phase). The eigenspectra of the relevant non-local operators are determined by the algebraic properties of the input group, G . For a general Abelian group we showed the ground state is $|G|^2$ -fold degenerate and outlined how to calculate ground state degeneracies associated to a non-Abelian input groups (using S_3 as an example). Finally, we related the representation theory of the quantum double algebra (which classifies the model's anyonic content) to the representation theory of the input group's centralisers. The anyonic content of $D(\mathbb{Z}_2)$ and $D(S_3)$ were computed as examples, with $D(S_3)$ giving rise to non-Abelian anyons.

Besides quantum double models, which are parametrised by a finite group G , one can study so-called 'string nets' which are parametrised by unitary fusion categories. String net models may give rise to non-Abelian anyons with non-integer quantum dimension, which is beyond the reach of any quantum double model. As an example, the Fibonacci anyons encountered in §2 have irrational quantum dimension and are realised as the quasiparticle excitations of a string net model [13]. String nets therefore provide platform for studying non-Abelian anyons which is strictly broader than the more restrictive quantum double framework (which exists as a proper subset of all string net models).

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